

Understanding the SymPy Code for 2-Qubit Quantum Channel Discovery

1. Objective of the Program

The purpose of this program is to **discover an unknown 4-qubit entangled quantum channel** that perfectly teleports an arbitrary **2-qubit quantum state** using a **fixed Bell measurement**.

Instead of assuming a channel beforehand, every channel coefficient is treated as an unknown variable and solved symbolically.

2. Importing Required Functions

```
from sympy import symbols, Matrix, kronecker_product, solve, sqrt
```

These functions are used as follows:

Function	Purpose
<code>symbols</code>	Creates symbolic variables.
<code>Matrix</code>	Represents vectors and matrices.
<code>kronecker_product</code>	Computes tensor products.
<code>solve</code>	Solves symbolic equations.
<code>sqrt</code>	Represents exact square roots.

3. Computational Basis States

```
q0 = Matrix([1,0])  
q1 = Matrix([0,1])
```

These are the computational basis vectors

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Every qubit in the program is constructed from these two vectors.

4. Arbitrary Two-Qubit Input

```
alpha, beta, gamma, delta = symbols('alpha beta gamma delta')
```

The four coefficients are symbolic.

The arbitrary input state is

$$|\psi\rangle = \alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle.$$

The program writes this as

```
input_state =  
alpha * kronecker_product(q0,q0)  
+  
beta * kronecker_product(q0,q1)  
+  
gamma * kronecker_product(q1,q0)  
+  
delta * kronecker_product(q1,q1)
```

Each tensor product gives

$$\begin{aligned} |00\rangle &= \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, & |01\rangle &= \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \\ |10\rangle &= \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, & |11\rangle &= \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}. \end{aligned}$$

Therefore

$$|\psi\rangle = [\alpha \ \beta \ \gamma \ \delta].$$

Thus

```
input_state
```

is a 4×1 column vector.

5. Unknown Quantum Channel

```
c = symbols('c1:17')
unknown_channel = Matrix(c)
```

The notation

```
symbols('c1:17')
```

creates

$$c_1, c_2, \dots, c_{16}.$$

Hence

$$|\Gamma\rangle = \begin{bmatrix} c_1 & c_2 & \vdots & c_{16} \end{bmatrix}.$$

This is a completely unknown four-qubit state.

The objective of the program is to determine these sixteen coefficients.

6. Entire Six-Qubit System

The joint state is

```
total_system =
kronecker_product(input_state,
unknown_channel)
```

Mathematically,

$$|\Psi_{\text{total}}\rangle = |\psi\rangle \otimes |\Gamma\rangle.$$

Since

- input state has dimension 4,
- channel has dimension 16,

the total vector has dimension

$$4 \times 16 = 64.$$

Therefore

```
total_system
```

is a 64×1 column vector.

7. Constructing One Bell State

The Bell state used is

```
bell_single =  
(1/sqrt(2))  
*  
(  
|00\rangle  
+  
|11\rangle  
)
```

implemented as

```
bell_single =  
(1/sqrt(2))  
*  
(  
kronecker_product(q0,q0)  
+  
kronecker_product(q1,q1)  
)
```

This equals

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}} [1\ 0\ 0\ 1].$$

8. Bell Measurement on Four Qubits

Alice performs

$$|\Phi^+\rangle_{13} \otimes |\Phi^+\rangle_{24}.$$

This is **not**

$$|\Phi^+\rangle_{12} \otimes |\Phi^+\rangle_{34},$$

because Bell measurements pair

- Input qubit 1 with Channel qubit 1,
- Input qubit 2 with Channel qubit 2.

Hence the tensor ordering must be rearranged.

9. Why Four Nested Loops?

```
for i in range(2):
    for j in range(2):
        for k in range(2):
            for l in range(2):
```

Each variable takes values

0, 1.

Thus there are

$$2^4 = 16$$

possible basis states

$$|ijkl\rangle.$$

Each iteration constructs one component of the four-qubit Bell measurement vector.

10. Computing the Bell Amplitude

Inside the loop,

```
amp =
bell_single[i*2+k]
*
bell_single[j*2+l]
```

The Bell state has components

$$00 \rightarrow \frac{1}{\sqrt{2}}, 01 \rightarrow 0, 10 \rightarrow 0, 11 \rightarrow \frac{1}{\sqrt{2}}.$$

The indices

$$i * 2 + k$$

represent the pair

$$(i, k),$$

while

$$j * 2 + l$$

represent

$$(j, l).$$

Therefore

$$\text{amp} = \langle ik | \Phi^+ \rangle \langle jl | \Phi^+ \rangle.$$

11. Why the Index Formula?

The basis ordering is

$$|ijkl\rangle.$$

Binary numbering gives

$$\text{index} = 8i + 4j + 2k + l.$$

Hence

```
idx =  
i*8  
+  
j*4  
+  
k*2  
+  
l
```

stores the coefficient in the correct location.

For example,

State	Index
0000	0
0001	1
0010	2
0011	3
0100	4
1111	15

Thus

```
bell_4qubit[idx]=amp
```

constructs the complete Bell measurement vector.

12. Reshaping the Total System

The total state has size

$$64 \times 1.$$

The command

```
system_matrix =  
total_system.reshape(16,4)
```

creates

$$16 \times 4.$$

Conceptually,

Rows = Alice's basis,
Columns = Bob's basis.

Thus the six-qubit state is viewed as

$$16 \times 4.$$

13. Performing the Bell Projection

```
bob_leftover =
(
bell_4qubit.T
*
system_matrix
).T
```

Dimensions are

$$(1 \times 16)(16 \times 4) = 1 \times 4.$$

After transpose,

$$4 \times 1.$$

Mathematically,

$$|\psi_{\text{Bob}}\rangle = (\langle\Phi_{13}^+| \otimes \langle\Phi_{24}^+|) |\Psi_{\text{total}}\rangle.$$

Thus the Bell measurement projects Alice's qubits and leaves Bob's two-qubit state.

14. Desired Teleported State

The target is

```
target_state =
Matrix([
alpha,
beta,
gamma,
delta
])
```

which represents

$$|\psi\rangle = [\alpha \ \beta \ \gamma \ \delta].$$

Bob should receive exactly this state.

15. Generating Symbolic Equations

The code

```
for r in range(4):  
    for param in  
        [  
            alpha,  
            beta,  
            gamma,  
            delta  
        ]:
```

extracts the coefficient of each independent parameter.

For example,

if

$$Bob_1 = \alpha c_1 + \beta c_2 + \gamma c_3 + \delta c_4,$$

then

```
Bob_1.coeff(alpha)
```

returns

$$c_1.$$

Similarly,

```
Bob_1.coeff(beta)
```

returns

$$c_2.$$

This produces

$$4 \times 4 = 16$$

linear equations.

16. Solving the Channel

Finally,

```
solution =  
solve(  
equations,  
c  
)
```

asks SymPy to determine

$$c_1, c_2, \dots, c_{16}$$

so that

$$|\psi_{\text{Bob}}\rangle = |\psi\rangle$$

for every

$$\alpha, \beta, \gamma, \delta.$$

Thus the unknown channel is obtained symbolically.

17. Final Output

```
print(solution)
```

prints the symbolic values of

$$c_1, c_2, \dots, c_{16}.$$

These coefficients describe every four-qubit entangled channel that allows perfect teleportation under the chosen Bell measurement. If the returned solution is proportional to a state vector, an additional normalization step is required to obtain a valid quantum state.